Multiscale Deep Learning Network With Self-Calibrated Convolution for Hyperspectral and LiDAR Data Collaborative Classification

Zhixiang Xue, Xuchu Yu, Xiong Tan, Bing Liu, Anzhu Yu, and Xiangpo Wei

***Abstract*— In this article, we propose a novel multiscale deep learning network with self-calibrated convolution (MSNetSC) for hyperspectral and light detection and ranging (LiDAR) data collaborative classification. Conventional deep learning methods have limitations in extracting multiscale features at a granular level from multimodality data and fusing these features in a context-awareness way, which will severely restrict the performance of hyperspectral and LiDAR data joint classification. The proposed multiscale deep learning network utilizes a hierarchical residual structure combined with self-calibrated convolution to extract features with different receptive fields, and this can enhance the model's capability to represent the multimodality data. Besides, we employ spectral and spatial self-attention modules to adaptively calibrate weights of features with different scales, thereby enhancing the discriminative ability of extracted multiscale features. Furthermore, the attentional feature fusion module can dynamically and adaptively fuse the features from multimodality data in a contextual scale-aware way, and this attention-based feature fusion method will further improve the collaborative classification performance of hyperspectral and LiDAR data. Four benchmark multimodality data (i.e., hyperspectral and LiDAR data) sets collected by different sensors and at different acquisition times are employed for joint classification experiments. These comparative classification results and ablation study sufficiently certify the superiority of the proposed model in terms of collaborative classification accuracy when compared with other state-of-the-art methods.**

***Index Terms*— Attentional feature fusion (AFF), collaborative classification, hierarchical residual network, hyperspectral image (HSI), light detection and ranging (LiDAR), self-calibrated convolution.**

I. Introduction

ACCURATE land-cover classification of remote sensing images is one of the most important components in the Earth observation (EO) missions. Containing rich spectral and spatial information simultaneously, hyperspectral images (HSIs) have unique advantages for the land-cover accurate recognition and classification, which have been widely applied in, e.g., resource management, urban planning, forest monitoring, military, and security applications [1], [2].

However, for many urban and rural scenes, HSIs have difficulties in discriminating complex objects having similar spectral characteristics. For example, the building roof and road are both composed of concrete thus having approximately the same spectral property, but they represent two different ground objects. Different from HSIs, light detection and ranging (LiDAR) data have the potential to record the elevation and structure of different surface targets, which can provide valuable complementary information for hyperspectral data land-cover classification. On the other hand, it is very difficult to distinguish ground objects (e.g., concrete and asphalt) with the same height but different materials only using the LiDAR data. Therefore, it is a promising paradigm to utilize multimodality data (e.g., hyperspectral and LiDAR data) for collaborative land-cover classification [3].

Many methods have been proposed to simultaneously utilize hyperspectral and LiDAR data for joint classification. Among these methods, one typical class of models is the feature-based fusion framework. The attribute profiles (APs) can extract spatial features from his- and LiDAR-derived digital surface model (DSM), and then, support vector machine (SVM), random forest (RF), and multinomial logistic regression (MLRsub) utilize stacked spatial and spectral features for joint classification [4]–[6]. Another type of spatial feature extraction method is the extinction profiles (EPs), which can extract spatial and elevation information from hyperspectral and LiDAR-derived DSM, afterward the composite kernel-based SVM, sparse, and low-rank component analysis (SLRCA) and total variation component analysis (OTVCA) fuse multiple features for final classification [7]–[9]. In [10], the morphological filters are applied to model the height and spatial information from the LiDAR-derived DSM and first few components of hyperspectral data, respectively. Then, the semisupervised graph fusion (SSGF) projects three types of features into a lower subspace to fuse these features for final supervised classification. In the multiple feature-based superpixel-level decision fusion (MFSuDF), the superpixel-guided kernel principal component analysis (KPCA) is applied for dimension reduction and noise suppression; then, 2-D and 3-D Gabor filters are utilized to obtain Gabor features from hyperspectral and LiDAR data, and after that, RF classifier and quadrant bit coding (QBC) accomplish the final classification task [11]. The learnable manifold alignment (LeMA) is a semisupervised cross-modality learning framework for hyperspectral and multispectral data, which can learn a joint graph structure to capture the data distribution for a more accurate decision boundary [12].

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Another type of feature fusion classification method is the deep learning-based architecture. Containing multiple processing layers, the deep learning models can learn abstract, intricate, and discriminative features from raw data, and these features can be fused for multimodality data joint classification [13], [14]. In [15], the EPs are utilized to learn spatial and elevation information from hyperspectral and LiDAR data, then, concatenated features are fed into a convolutional neural network (CNN) to produce the final classification map, and this is the first time to employ the deep learning-based model for HSI and LiDAR joint classification. In [16], double-branch structure is first proposed to extract spectral–spatial features from HSI and contextual features from LiDAR data, respectively, and these heterogeneous features are concatenated for final joint classification. Also, under the double-branch structure, the CNN with cascade block and 3-D capsule network (CapsNet) are employed to extract multiple features from hyperspectral and LiDAR data for joint classification [17], [18]. In the dual-channel spatial, spectral, and multiscale attention convolutional long short-term memory neural network (dual-channel A^3CLNN), the composite attention learning mechanism combined with a three-level fusion module is employed to integrate multiple features from hyperspectral and LiDAR data [19]. In the framework of three-branch structure, the CNN and residual convolutional neural networks (ResNet) are exploited to extract spatial, spectral, and elevation information separately, and these features are concatenated for joint classification [20], [21].

The straightforward concatenation or summation of features from hyperspectral and LiDAR data will increase the dimension of features, which may cause the Hughes phenomenon and thus affect the joint classification accuracy. In [22], two coupled CNNs using parameter-sharing strategy are employed to learn spectral–spatial features and elevation information from hyperspectral and LiDAR data, respectively, and these heterogeneous features are fused via feature- and decision-level fusion strategies. The encoder–decoder architecture (EndNet) exploits the encoder network to extract the features from multimodality data and fuses the multimodal information by enforcing the fused features to reconstruct the multimodel input [23]. The end-to-end multimodel deep learning framework employs extraction network and fusion network to conduct the cross-modality classification [24]. Also, the patch-to-patch convolutional neural network (PToP CNN) uses the encoder–decoder translation architecture to learn joint features between hyperspectral and LiDAR data in an unsupervised way. Then, different features derived from multiple hidden layers of this network are fused by the hierarchical fusion module and stacked into a vector for final classification [25]. In the hierarchical random walk network (HRWN), the spectral and spatial features are extracted by a dual-tunnel CNN. Then, in the hierarchical random walk layer, the predicted distribution of dual-tunnel CNN serves as global prior, while pixelwise affinity reflects the local similarity of pixel prior, which enforces spatial consistency in the deeper layers of networks [26]. The feature fusion and extraction framework (FusAtNet) utilizes a self-attention mechanism to highlight the extracted features, and a cross-attention approach is also used to harness the LiDAR-derived attention map that accentuates the spatial features of HSI for the land-cover classification [27].

Nevertheless, there still exist some issues in hyperspectral and LiDAR collaborative classification. It is a fact that current classification methods have limitations in extracting multiscale features from multimodality datasets, and these features are vital for the representation of multimodality data, especially for complex data. Also, features of different scales make unequal contributions to classification performance, which needs to be adaptively calibrated. What is more, fusing multimodality features by simple concatenation may further increase the dimensionality, and due to lack of large labeled data, the model may suffer from curse of dimensionality and affect final multimodality data classification accuracy. All the above factors will affect the hyperspectral and LiDAR data collaborative classification performance to a certain extent.

In the remote sensing community, there are different modalities of remote sensing data within the same scene, and the observed scenes contain many types of land-cover classes. Aiming to solve the problems specific in remote sensing classification, we utilize a variety of strategies to comprehensively use multimodality remote sensing data for collaborative classification. Specifically, the multiscale deep learning network with self-calibrated convolution (MSNetSC) not only extracts spectral, spatial, and elevation features of different scales but also employs a self-attention mechanism to promote the discriminative ability of extracted multiscale features. Also, the nonlinear attentional feature fusion (AFF) module can dynamically and adaptively fuse the features from multimodality data in a contextual scale-aware way for better hyperspectral and LiDAR data joint classification. More specifically, the main contributions of our article can be summarized as follows.

1) First, the hierarchical residual structure combined with self-calibrated convolution can extract spatial, spectral, and elevation information of different receptive fields, and these multiscale features have strong representation capability for multimodality data, especially for more complex dataset.
2) Second, spectral and spatial self-attention modules specific to heterogeneous features are employed to explicitly calibrate the weights of features of different scales, which can further improve the discriminative ability of extracted multiscale features.
3) Third, the AFF module can adaptively fuse the spectral, spatial, and elevation features in a contextual scale-aware way, and this nonlinear attention-based feature fusion method will improve multimodality data classification performance to a certain degree, which also provides a novel feature fusion mode for other remote sensing data.
4) The ablation study and comparative classification experiments conducted on four different multimodality datasets certify the effectiveness and advancement of the proposed method.

The structure of this article is organized as follows. The proposed MSNetSC is described in detail in Section II. Section III reports experimental setup, comparative classification results,

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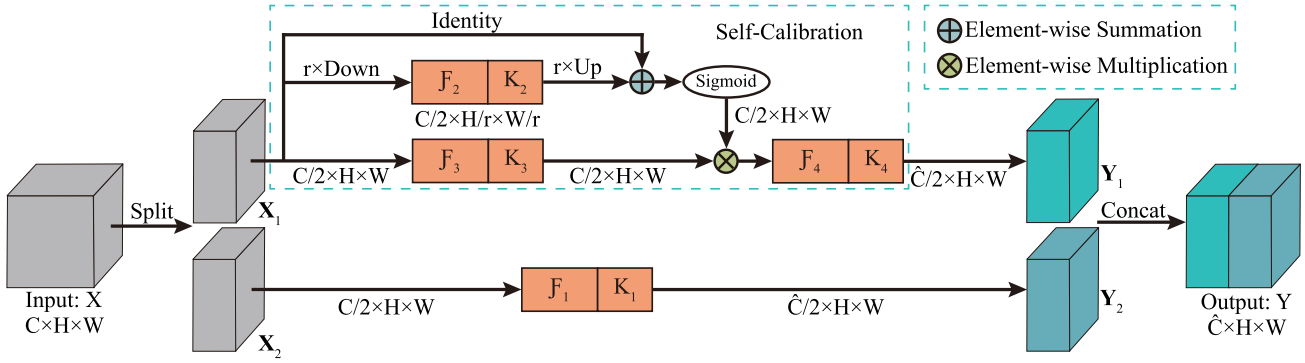

Fig. 1. Illustration of the self-calibrated convolution. The input $\boldsymbol{X}$ is split into two portions $\{\boldsymbol{X}_1, \boldsymbol{X}_2\}$, and then, they perform the self-calibration operation and conventional convolution operation, respectively. Also, their intermediate outputs $\{\boldsymbol{Y}_1, \boldsymbol{Y}_2\}$ are concatenated together as $\boldsymbol{Y}$ to collect different types of contextual information.

parameter analysis, and ablation study. Finally, Section IV concludes this article.

II. Methodology

The proposed MSNetSC simultaneously utilizes the representation ability of multiscale features and the weight recalibration capability of attention mechanism for hyperspectral and LiDAR data collaborative classification. First, the hierarchical residual structure combined with self-calibrated convolution not only improves the receptive field within the convolutional layer but also obtains different scale receptive fields of the entire feature map. Thus, the extracted features with different receptive fields can represent the multiscale information of multimodality data. Then, three self-attention modules, corresponding to different types of features, adaptively assign optimal weights for features of different scales. Finally, the AFF module dynamically assigns the corresponding weights for different modality features during the training process, which effectively fuses multimodality features for collaborative classification.

A. Self-Calibrated Convolution

In the traditional convolutional layer $\mathcal{F}$, the field-of-view for each spatial location in the feature map is restricted by predefined kernel size because the output feature map is computed by summation in a local region with the same pattern. Also, networks composed of such convolutional layers are lack of large enough receptive fields, and thus, they cannot capture high-level semantics [28]. However, in the self-calibrated convolutional layer, the feature transformation is conducted in two different scale spaces, and the latent space after down-sampling operation can guide the original feature transformation to generate more expressive feature representations [29]. More specifically, the latent space after down-sampling has a larger field-of-view than the original space, and the embeddings after transformation in the latent space can be employed as a reference to guide the feature transformation in the original space. The schematic illustration of the self-calibrated convolution is shown in Fig. 1.

In the self-calibrated convolution operation, the input $\boldsymbol{X}$ is split into two portions $\{\boldsymbol{X}_1, \boldsymbol{X}_2\}$, and they carry out different types of operations. There are four convolution operations, denoted as $\{\boldsymbol{K}_1, \boldsymbol{K}_2, \boldsymbol{K}_3, \boldsymbol{K}_4\}$. The $\boldsymbol{X}_2$ portion performs the traditional convolution operation $\boldsymbol{Y}_2 = \mathcal{F}_1(\boldsymbol{X}_2) = \boldsymbol{X}_2 * \boldsymbol{K}_1$, with a view to retain the original spatial context. Also, another portion $\boldsymbol{X}_1$ implements the self-calibration operation.

For $\boldsymbol{X}_1$, we first conduct down-sampling operation using the average pooling with size r as follows:

$$\boldsymbol{T}_1 = \text{AvgPool}_r(\boldsymbol{X}_1). \tag{1}$$

Also, the feature transformations on $\boldsymbol{T}_1$ are performed with $\boldsymbol{K}_2$, which can be denoted as

$$\boldsymbol{X}_1' = \text{Up}(\mathcal{F}(\boldsymbol{T}_1)) = \text{Up}(\boldsymbol{T}_1 * \boldsymbol{K}_2) \tag{2}$$

in which $\text{Up}(\cdot)$ is a bilinear interpolation operator, which transforms the features from latent space to the original feature space. Then, the calibration operation is formulated as follows:

$$\boldsymbol{Y}_1' = \mathcal{F}_3(\boldsymbol{X}_1) \cdot \sigma\left(\boldsymbol{X}_1 + \boldsymbol{X}_1'\right) \tag{3}$$

where $\mathcal{F}_3(\boldsymbol{X}_1) = \boldsymbol{X}_1 * \boldsymbol{K}_3$, σ is the sigmoid function, and $\cdot$ is the element-wise multiplication. In addition, $\boldsymbol{X}_1'$ is employed as residual to form weights for calibration. Thus, the final self-calibration operation can be written as follows:

$$\boldsymbol{Y}_1 = \mathcal{F}_4\left(\boldsymbol{Y}_1'\right) = \boldsymbol{Y}_1' * \boldsymbol{K}_4. \tag{4}$$

Also, the outputs $\{\boldsymbol{Y}_1, \boldsymbol{Y}_2\}$ after performing different operations are concatenated as $\boldsymbol{Y}$ to collect different types of contextual information. From the self-calibrated convolution operation, we can observe that each spatial location not only adaptively considers its surrounding informative context as embeddings from the latent space as guidance for the original scale space but also models interchannel dependencies. What is more, the self-calibration operation only considers an appropriate range of context for each spatial location, which can also avoid some contaminating information from the irrelevant region. Thus, the self-calibrated convolution can extract more discriminative features by expanding the field-of-view within the self-calibrated convolutional layer.

B. Hierarchical Residual Learning

In the task of visual recognition, the features of objects show a multiscale morphology, and thus, multiscale feature

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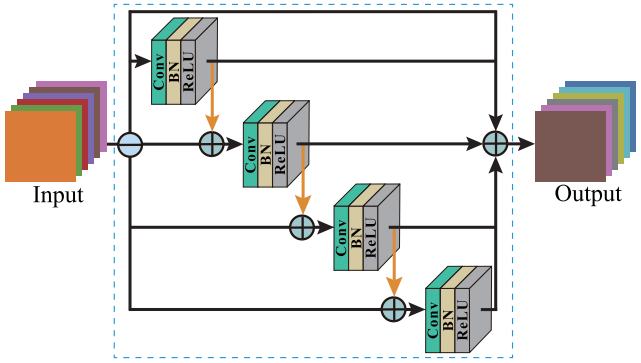

Fig. 2. Illustration of the hierarchical residual unit, in which the scale factor is five. Conv denotes the convolutional layer, BN denotes the batch normalization layer, and ReLU denotes the rectified linear unit activation function.

representation is necessary for image classification. Conventional CNNs obtain multiscale features by stacking a different number of convolutional layers, but these features have relatively fixed receptive fields. Similar to the residual network in structure, the hierarchical residual learning introduces a new scale dimension as an essential factor except for existing dimensions of depth, width, and cardinality [30]. Also, the scale dimension in hierarchical residual network means the number of subgroups in the hierarchical residual unit. Because different subgroups of feature maps have different receptive fields, the combined feature maps represent features of multiple receptive fields, and therefore, the hierarchical residual unit can represent multiscale information at a granular level [31]. Fig. 2 shows a hierarchical residual unit with five scales, in which $\ominus$ and $\oplus$ represent the split operation and concatenation operation, respectively.

We denote the input and output of the hierarchical residual unit as $\boldsymbol{X}$ and $\boldsymbol{Y}$, respectively. The input feature maps $\boldsymbol{X}$ are first split into s feature subsets, s denotes scale factor, and every subset is represented as $\boldsymbol{X}_i$, where $i \in \{1, 2, \ldots, s\}$. The subset $\boldsymbol{X}_i$ has the same spatial size with input $\boldsymbol{X}$, but only $1/s$ channels. Except for $\boldsymbol{X}_1$, every $\boldsymbol{X}_i$ has corresponding convolution operations, denoted with $\boldsymbol{K}_i()$. Also, we use $\boldsymbol{Y}_i$ to denote the output of $\boldsymbol{K}_i()$. To obtain hierarchical features, we add the output of $\boldsymbol{K}_{i-1}()$ to the feature subset $\boldsymbol{X}_i$, and then, they are fed into $\boldsymbol{K}_i()$. Thus, $\boldsymbol{Y}_i$ can be generally written as follows:

$$\boldsymbol{Y}_i = \begin{cases} \boldsymbol{X}_i, & i = 1; \\ \boldsymbol{K}_i(\boldsymbol{X}_i), & i = 2; \\ \boldsymbol{K}_i(\boldsymbol{X}_i + \boldsymbol{Y}_{i-1}), & 2 < i \leq s. \end{cases} \tag{5}$$

In the hierarchical residual unit, each convolution operator $\boldsymbol{K}_i()$ can receive information from subsets $\boldsymbol{X}_j (j \leq i)$, and thus, the feature $\boldsymbol{X}_i$ has a larger receptive field than $\boldsymbol{X}_j$. Also, feature maps of different receptive fields are concatenated at the end of hierarchical residual unit to represent multiscale features. In the hierarchical residual network, larger scale factor s allows the unit to learn features with richer receptive fields. To normalize multiscale features for better classification performance, we add the batch normalization (BN) layer and rectified linear unit (ReLU) activation function after every convolutional layer. Also, multiscale spectral and spatial features are extracted by specific convolution kernels. Thus, the hierarchical residual network with residual-like connections could capture multiscale features at a granular level.

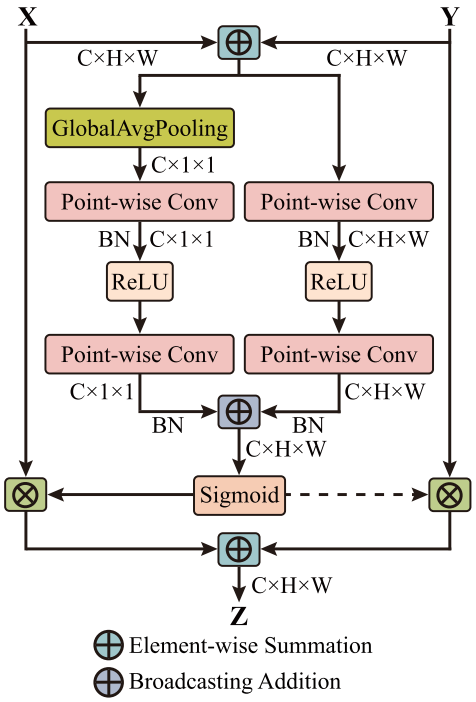

Fig. 3. Illustration of the AFF module. Point-wise Conv denotes the point-wise convolutional layer and GlobalAvgPooling denotes the global average pooling layer.

C. Attentional Feature Fusion

The linear feature fusion approaches (e.g., addition and concatenation) are entirely contextual unaware, which merely offer a fixed linear aggregation of feature maps, and these fusion methods are unaware of whether this combination is suitable for specific purposes. Inspired by the visual perception of the human visual system, the attention mechanism assigns the corresponding weights according to the importance of the targets [32]. The AFF method presents a nonlinear approach for feature fusion, which dynamically and adaptively fuses the received features in a contextual scale-aware way for adaptive feature fusion [33]. Also, the schematic illustration of the AFF module is shown in Fig. 3.

Given two features $\boldsymbol{X}, \boldsymbol{Y} \in \mathbb{R}^{C \times H \times W}$ to be fused, the initial integration $\boldsymbol{F}$ of two types of features is conducted by the element-wise summation operation. Also, the point-wise convolution (PWConv) is selected as the local channel context aggregator, which can achieve cross-channel information fusion for each spatial position. In this fusion module, the local channel context and the global channel context are obtained through different series of operations. More specifically, the local channel context $\boldsymbol{L}(\boldsymbol{M}) \in \mathbb{R}^{C \times H \times W}$ is computed via a bottleneck structure as follows:

$$\boldsymbol{L}(\boldsymbol{F}) = \mathcal{B}(\mathrm{PWConv}_2(\delta(\mathcal{B}(\mathrm{PWConv}_1(\boldsymbol{F}))))). \tag{6}$$

Also, the global channel context $\boldsymbol{G}(\boldsymbol{F})$ is calculated as follows:

$$\boldsymbol{G}(\boldsymbol{F}) = \mathcal{B}(\mathrm{PWConv}_4(\delta(\mathcal{B}(\mathrm{PWConv}_3(g(\boldsymbol{F})))))) \tag{7}$$

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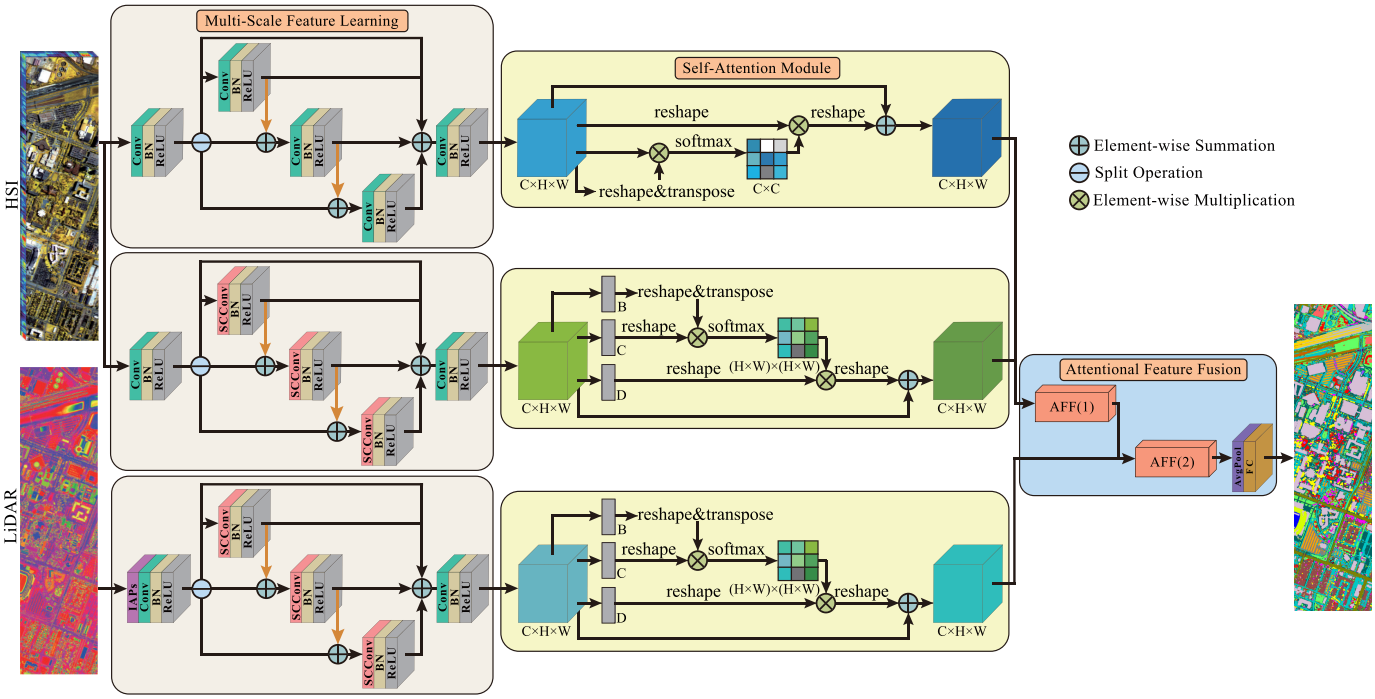

Fig. 4. Schematic framework of MSNetSC for hyperspectral and LiDAR data collaborative classification (the representative MSNetSC model has four scales and six blocks). SCConv denotes a novel type of convolution, which is the self-calibrated convolution, AvgPool denotes the adaptive average pooling layer, FC denotes the fully connected layer, and IAPs means the IAPs.

where $g(\boldsymbol{F})$ denotes the global average pooling operation, δ denotes the ReLU activation function, and $\mathcal{B}$ denotes the BN. It is noteworthy that $\boldsymbol{L}(\boldsymbol{F})$ has the same shape with the initial feature $\boldsymbol{F}$, which can preserve and highlight the subtle details in the low-level features. Then, the attentional weights generated by the AFF module is calculated by

$$M(\boldsymbol{F}) = \sigma(L(\boldsymbol{F}) \oplus \boldsymbol{G}(\boldsymbol{F})). \tag{8}$$

Thus, the AFF can be expressed as

$$\boldsymbol{Z} = M(\boldsymbol{F}) \otimes \boldsymbol{X} + (1 - M(\boldsymbol{F})) \otimes \boldsymbol{Y} \tag{9}$$

where $\boldsymbol{Z} \in \mathbb{R}^{C \times H \times W}$ is the final fused feature maps and $\oplus$ and $\otimes$ denote the broadcasting addition operation and element-wise multiplication operation, respectively. In this fusion module, the fusion weights $M(\boldsymbol{F})$ consist of real numbers between 0 and 1, so are $1 - M(\boldsymbol{F})$, which enables the network to conduct a soft selection or weighted averaging between $\boldsymbol{X}$ and $\boldsymbol{Y}$. Also, the fusion weights are dynamically optimized during the training process, which ensures that two types of features are effectively fused.

D. Network Architecture of MSNetSC

The whole framework of MSNetSC is shown in Fig. 4. In order to make the most of spectral, spatial, and elevation information of hyperspectral and LiDAR data, we adopt the three-branch structure for multimodality data collaborative classification, and three branches represent spectral, spatial, and elevation branches. Also, the whole joint classification process includes three stages, namely multiscale feature learning, self-attentional feature recalibration, and AFF stages. For the purpose of focusing more on discriminative and effective features while weakening information detrimental to classification, we utilize self-attention modules to calibrate multiscale features in the feature recalibration stage. Also, there are two types of self-attention modules, in which the spectral attention module models the interdependencies between channels and the spatial attention module establishes the relationships in spatial domain to selectively emphasize informative pixels.

In the upper spectral branch, there are spectral-based hierarchical residual network and the corresponding spectral attention module. The spectral-based hierarchical residual network utilizes spectral-based convolution kernels to extract multiscale spectral features, and the spectral attention module recalibrates channel-wise features by explicitly establishing the relationships between channels. The middle spatial branch is composed of spatial-based hierarchical residual network and corresponding spatial attention module. This multiscale feature extraction network employs the self-calibrated convolution to extract more discriminative multiscale features, and the spatial attention module is used to selectively emphasize informative positions in the spatial domain. For similar purpose, the lower elevation branch simultaneously utilizes a hierarchical spatial residual network with self-calibrated convolution and corresponding spatial attention block. In order to fully exploit the useful information of the LiDAR data, we employ the invariant attribute profiles (IAPs) [34] to extend the single band of LiDAR-derived DSM to 21-band profiles. In the feature fusion stage, there are two AFF modules. The first AFF module is employed to fuse spectral and spatial features from hyperspectral data, and the second AFF module combines the features from different sources (i.e., hyperspectral and LiDAR data). During the training process of MSNetSC, self-attention modules and AFF modules adaptively select optimal

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TABLE I
DETAILED STRUCTURE AND PARAMETERS OF MSNETSC MODEL

Branch	Layer Name	Output Shape	Filter Size	Padding
Spectral	Input	7×7×63	N/A	N
	Conv1	7×7×30(24)	1×1×5(24)	N
	Conv21	7×7×30(6)	1×1×5(6)	Y
	Conv22	7×7×30(12)	1×1×5(12)	Y
	Conv23	7×7×30(12)	1×1×5(12)	Y
	Conv3	7×7×1(24)	1×1×30(24)	N
Spatial	Input	7×7×63	N/A	N
	Conv1	7×7×1(24)	1×1×63(24)	N
	SCConv21	7×7×1(6)	3×3×1(6)	Y
	SCConv22	7×7×1(12)	3×3×1(12)	Y
	SCConv23	7×7×1(12)	3×3×1(12)	Y
	Conv3	7×7×1(24)	1×1×1(24)	Y
LiDAR	Input	7×7×21	N/A	N
	Conv1	7×7×1(24)	1×1×21(24)	N
	SCConv21	7×7×1(6)	3×3×1(6)	Y
	SCConv22	7×7×1(12)	3×3×1(12)	Y
	SCConv23	7×7×1(12)	3×3×1(12)	Y
	Conv3	7×7×1(24)	1×1×1(24)	Y
Fusion	Linear	24	N/A	N
	Output	6	N/A	N

weights for multiscale features and multimodality features, respectively.

In order to simultaneously utilize multimodality features for collaborative classification, we employ different branches to extract specific features, and the inputs of three branches are HSI block and IAPs block. Since these heterogeneous features come from different modal data, to maintain the balance between these features as much as possible, we employ the same number of feature extraction layers. Also, we further normalize these features in the feature fusion stage to further adjust the features for fusion. The schematic model in Fig. 4 is an MSNetSC model with four scales and six blocks, and the corresponding detailed parameters are listed in Table I. The Trento dataset is employed in this representative model, and the spatial size is set as 7. In our proposed MSNetSC method, we employ the convolution kernel with (1, 1, 5) and self-calibrated convolution kernel with (3, 3, 1) to extract multiscale spectral and spatial features, respectively. Note that the stride of Conv1 in the spectral branch is (1, 1, 2) and the stride of other convolution and self-calibrated convolution in MSNetSC is (1, 1, 1).

III. EXPERIMENTS AND ANALYSIS

All collaborative classification experiments are carried out on a workstation equipped with an Intel Core i9-7900X, an Nvidia Geforce GTX 2080 Ti GPU, and 128-GB RAM. The proposed MSNetSC model is implemented with the PyTorch library using Python language, and other comparative methods use the original corresponding experiment environment.

A. Datasets Description

In order to assess the effectiveness and advancement of the proposed method, four benchmark multimodality data (i.e., hyperspectral and LiDAR data) sets are employed to conduct the comparative classification experiments. For a more fair comparison, we employ the same specific number of training samples with other state-of-the-art methods in all classification experiments. The brief description of four benchmark datasets is given as follows.

TABLE II
LAND-COVER CLASSES AND SAMPLES OF THE TRENTO DATASET

Class No.	Class Name	Training	Test	Percent
1	Apple Trees	129	3905	3.20%
2	Buildings	125	2778	4.31%
3	Ground	105	374	21.92%
4	Wood	154	8969	1.69%
5	Vineyard	184	10317	1.75%
6	Roads	122	3052	3.84%
	Total	819	29395	2.71%

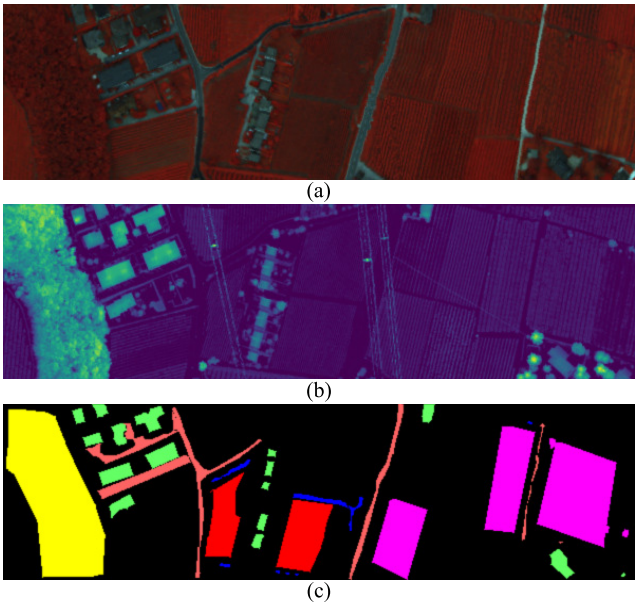

Fig. 5. Visualization of the Trento dataset. (a) Pseudo-color image of the hyperspectral data. (b) Grayscale image of the LiDAR-derived rasterized DSM. (c) Ground-truth map.

1) Trento: The Trento dataset covers a rural area in the south of Trento, Italy. The hyperspectral data were acquired by the AISA Eagle sensor, which consists of 63 bands ranging from 402.89 to 989.09 nm, and the spectral resolution is 9.2 nm. The corresponding LiDAR data were captured by the Optech Airborne Laser Terrain Mapper (ALTM) 3100EA sensor. The spatial size of this dataset is 166 × 600 pixels, and the spatial resolution is 1 m. There are six land-cover classes in this dataset, and the number of training and test samples is listed in Table II. Also, Fig. 5 shows the visualization of Trento dataset.

2) MUUFL Gulfport: The MUUFL Gulfport dataset was collected over the University of Southern Mississippi Gulf Park campus, Long Beach, MS, USA, in November 2010. The hyperspectral data were acquired by the ITRES Compact Airborne Spectrographic Imager (CASI-1500) sensor, which covers the wavelength range from 367.7 to 1043.4 nm with the spectral resolution of 9.5–9.6 nm [35], [36]. Eight bands are removed from the original hyperspectral data because of

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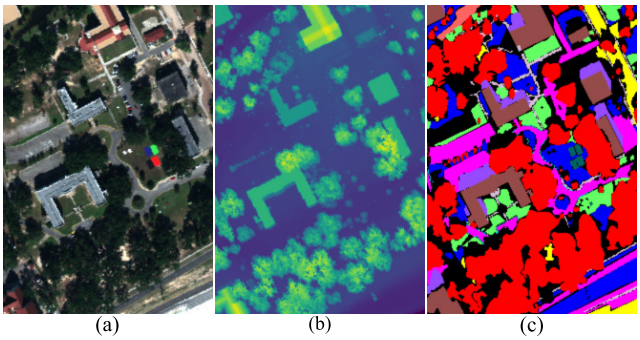
TABLE III

LAND-COVER CLASSES AND SAMPLES OF THE MUUFL GULFPORT DATASET

Class No.	Class Name	Training	Test	Percent
1	Trees	150	23096	0.65%
2	Mostly Grass	150	4120	3.51%
3	Mixed Ground Surface	150	6732	2.18%
4	Dirt and Sand	150	1676	8.21%
5	Road	150	6537	2.24%
6	Water	150	316	32.19%
7	Building Shadow	150	2083	6.72%
8	Buildings	150	6090	2.40%
9	Sidewalk	150	1235	10.83%
10	Yellow Curb	100	83	54.64%
11	Cloth Panels	100	169	37.17%
	Total	1550	52137	2.89%



(a) (b) (c)

Fig. 6. Visualization of the MUUFL Gulfport dataset. (a) Pseudo-color image of the hyperspectral data. (b) Grayscale image of the LiDAR-derived rasterized DSM. (c) Ground-truth map.

noise, and the remaining 64 bands are utilized for experiments. Due to the invalid area in the original image, this HSI has also been spatially cropped into 325 × 220 pixels. The coregistered LiDAR data were captured by the ALTM sensor relying on a laser with the wavelength of 1064 nm. The MUUFL Gulfport dataset covers 11 classes of interests, and the corresponding number of training and test samples is listed in Table III. Also, Fig. 6 shows the visualization of MUUFL Gulfport dataset.

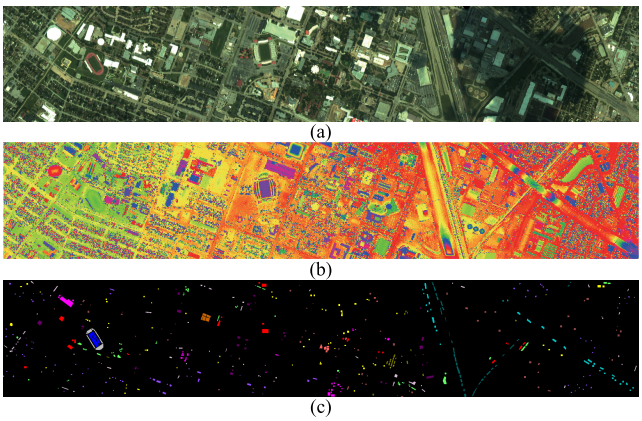
3) Houston 2013: The Houston 2013 dataset was collected by the NSF-funded National Center for Airborne Laser Mapping (NCALM) over the Houston University campus and surroundings in June 2012, which is provided by the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Competition [37]. This dataset contains hyperspectral imagery and corresponding coregistered LiDAR-derived DSM. There are 144 spectral bands within the wavelength range from 380 to 1050 nm in the HSI. Also, the spatial size of this dataset is 349 × 1905 pixels with the spatial resolution of 2.5 m. There are 15 distinguishable land-cover classes within the image coverage, and the detailed class names, training samples, as well as test samples are shown in Table IV. The false-color image of hyperspectral data, grayscale image of LiDAR-derived DSM, as well as ground-truth map are shown in Fig. 7.

4) Houston 2018: The Houston 2018 dataset was gathered over the University of Houston campus in February 2017,

TABLE IV

LAND-COVER CLASSES AND SAMPLES OF THE HOUSTON 2013 DATASET

Class No.	Class Name	Training	Test	Percent
1	Healthy grass	198	1053	15.83%
2	Stressed grass	190	1064	15.15%
3	Synthetic grass	192	505	27.55%
4	Trees	188	1056	15.11%
5	Soil	186	1056	14.98%
6	Water	182	143	56.00%
7	Residential	196	1072	15.46%
8	Commercial	191	1053	15.35%
9	Road	193	1059	15.42%
10	Highway	191	1036	15.57%
11	Railway	181	1054	14.66%
12	Parking lot 1	192	1041	15.57%
13	Parking lot 2	184	285	39.23%
14	Tennis court	181	247	42.29%
15	Running track	187	473	28.33%
	Total	2832	12197	18.84%



(a) (b) (c)

Fig. 7. Visualization of the Houston 2013 dataset. (a) Pseudo-color image of the hyperspectral data. (b) Grayscale image of the LiDAR-derived rasterized DSM. (c) Ground-truth map.

which is provided by the 2018 IEEE GRSS Data Fusion Competition [38]. The hyperspectral data were captured using the ITRES CASI-1500 sensor, and the corresponding LiDAR data were acquired by the Optech Titam multi-wave (MW) (14SEN/CON340) sensor. We only use the training portion of the whole hyperspectral data, and the corresponding LiDAR-derived rasterized DSM and ground truth are resampled to adapt the hyperspectral data [39]. After the normalization of spatial resolution, this dataset has the spatial size of 601 × 2384 pixels with 1-m ground sampling distance (GSD). Also, there are 48 spectral bands within the wavelength range from 380 to 1050 nm in the hyperspectral data. This dataset covers 20 urban land-cover classes, and the detailed number of training samples and test samples as well as the corresponding class names are shown in Table V. Also, Fig. 8 shows the illustration of pseudo-color image of hyperspectral data, grayscale image of LiDAR-derived rasterized DSM, as well as the ground-truth map.

B. Experimental Setup

The multimodality data classification performance is evaluated both quantitatively and qualitatively in our comparative

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TABLE V
LAND-COVER CLASSES AND SAMPLES OF THE HOUSTON 2018 DATASET

Class No.	Class Name	Training	Test	Percent
1	Healthy grass	1458	8341	14.88%
2	Stressed grass	4316	28186	13.28%
3	Artificial turf	331	353	48.39%
4	Evergreen trees	2005	11583	14.76%
5	Deciduous trees	676	4372	13.39%
6	Bare earth	1757	2759	38.91%
7	Water	147	119	55.26%
8	Residential buildings	3809	35953	9.58%
9	Non-residential buildings	2789	220895	1.25%
10	Roads	3188	42622	6.96%
11	Sidewalks	2699	31303	7.94%
12	Crosswalks	225	1291	14.84%
13	Major thoroughfares	5193	41165	11.20%
14	Highways	700	9149	7.11%
15	Railways	1224	5713	17.64%
16	Paved parking lots	1179	10296	10.27%
17	Unpaved parking lots	127	22	85.23%
18	Cars	848	5730	12.89%
19	Trains	493	4872	9.19%
20	Stadium seats	1313	5511	19.24%
	Total	34477	470235	6.83%

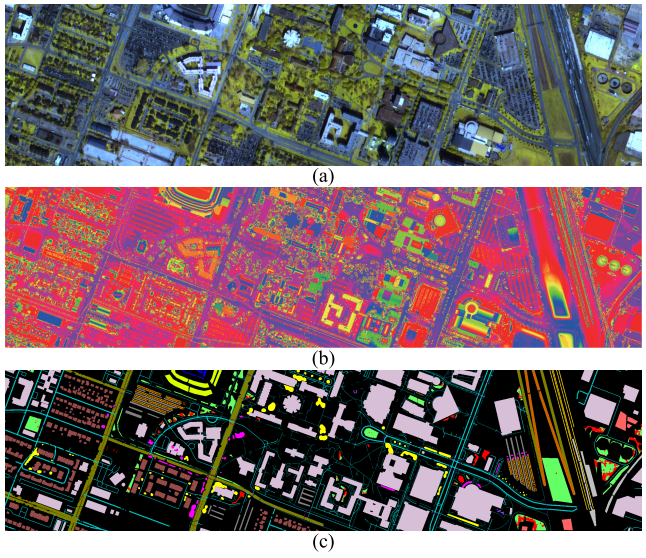

Fig. 8. Visualization of the Houston 2018 dataset. (a) Pseudo-color image of the hyperspectral data. (b) Grayscale image of the LiDAR-derived rasterized DSM. (c) Ground-truth map.

experiments. The main classification evaluation coefficients, namely overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and individual class accuracy, are utilized to assess the experimental results quantitatively. Classification maps are also exploited to evaluate the classification performance from the qualitative perspective.

In order to highlight the superiority of our proposed model in terms of classification performance, we employ several state-of-the-art methods for comparative experiments, which include four classification models originally for hyperspectral data and four multimodality data classification models. These classification models originally for hyperspectral data include the classic SVM with radial basis function (RBF) kernel in [40], the contextual deep convolutional neural network (CDCNN) in [41], the spectral–spatial residual network (SSRN) in [42], and the double-branch dual-attention mechanism network (DBDA) in [43]. Also, the multimodality data classification models include the two-branch convolutional neural network (TBCNN) in [17], the deep encoder–decoder network (EndNet) in [23], the multimodality learning framework using the cross fusion fully connected network (MML) in [24], and the HRWN in [26].

For a fair comparison, we stack the IAPs from LiDAR-derived DSM data with hyperspectral data as the input of these methods (i.e., SVM, CDCNN, SSRN, and DBDA) for multimodality data joint classification. In these original multimodality data classification models (i.e., TBCNN, EndNet, MML, and HRWN), we carry out classification experiments using the original network structure and parameters according to corresponding articles. We carry out ten trials for each classification experiment to increase the reliability and credibility, and we randomly select training samples in each trial. In each trial, 20% of training samples are set as a validation set, and the optimal trained model is selected by the classification accuracy on the validation set. Then, the corresponding hyperparameters are determined as optimal hyperparameters. Also, in each group of experiments, we utilize the same number of training samples for all classification methods.

C. Classification Results and Analysis

1) Quantitative Evaluation: The detailed classification results in terms of average OA, AA, kappa coefficient (κ) with the corresponding variance, and individual class accuracy obtained by different classification methods on four benchmark multimodality datasets are listed in Tables VI–IX, respectively. Note that the bold value in these tables represents the optimal value in the corresponding row. Several conclusions can be drawn from these tables.

First, using the same number of training samples, deeper models have higher classification accuracies. In four groups of classification experiments, the deeper models (e.g., SSRN, DBDA, and MSNetSC) generally have better classification performance than models with fewer layers (e.g., SVM, CDCNN, TBCNN, and EndNet). This can be explained by the fact that deeper models can extract more discriminative features for collaborative classification.

Second, multimodality data classification models (e.g., MML, HRWN, and MSNetSC) can obtain higher accuracies than classification models that simply concatenate features from hyperspectral and LiDAR data (e.g., SVM, CDCNN, SSRN, and DBDA). This is because different modality data have complementary information for classification, and the simple concatenation pattern cannot effectively fuse this useful information for collaborative classification.

Third, by introducing the self-attention mechanism, the deep learning models with self-attention modules (e.g., DBDA and MSNetSC) can improve the classification performance to a certain degree. The reason is that these self-attention modules can adaptively highlight informative features and suppress useless information during the training process. Thus, this feature calibration mechanism will improve the discriminative capability of features.

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TABLE VI

OA, AA, KAPPA COEFFICIENTS, AND CLASS-SPECIFIC ACCURACY(%) WITH DIFFERENT METHODS ON THE TRENTO DATASET

Class No.	SVM [40]	CDCNN [41]	SSRN [42]	DBDA [43]	TBCNN [17]	EndNet [23]	MML [24]	HRWN [26]	MSNetSC
1	69.75	97.91	99.96	98.06	72.29	73.10	96.01	96.19	**99.97**
2	95.96	95.39	88.76	96.11	96.86	96.29	88.64	**99.43**	98.28
3	88.31	96.71	98.75	98.97	94.22	89.53	98.02	**99.70**	97.74
4	99.61	99.85	99.93	99.88	99.95	99.56	99.68	99.65	**99.98**
5	95.73	99.66	99.90	98.96	99.88	96.96	99.43	99.40	**99.95**
6	95.58	97.31	96.33	98.30	**99.05**	96.76	93.51	94.87	98.91
OA	92.23	98.76	98.27	98.70	94.39	93.49	97.36	99.05	**99.66**
	0.67	0.72	1.22	0.81	0.61	0.45	0.17	0.18	**0.08**
AA	90.82	97.80	97.27	98.38	93.71	92.03	95.88	98.21	**99.14**
	1.06	0.99	1.35	0.82	0.56	0.38	0.40	0.20	**0.28**
κ	89.72	98.34	97.69	98.29	92.59	91.13	96.46	98.72	**99.55**
	0.87	0.96	1.64	1.09	0.65	0.47	0.23	0.25	**0.10**

TABLE VII

OA, AA, KAPPA COEFFICIENTS, AND CLASS-SPECIFIC ACCURACY(%) WITH DIFFERENT METHODS ON THE MUUFL GULFPORT DATASET

Class No.	SVM [40]	CDCNN [41]	SSRN [42]	DBDA [43]	TBCNN [17]	EndNet [23]	MML [24]	HRWN [26]	MSNetSC
1	98.28	98.87	**99.19**	99.06	98.98	97.66	98.45	93.86	98.98
2	66.56	72.18	77.87	77.17	68.79	69.00	71.30	**91.76**	84.63
3	83.35	82.89	**92.32**	90.27	87.37	82.86	84.23	87.50	90.63
4	80.35	80.00	88.72	89.31	80.03	82.34	82.96	**98.62**	91.32
5	94.24	**96.78**	96.75	96.06	92.58	93.59	95.60	93.99	95.86
6	65.56	54.68	87.02	86.05	77.25	73.05	70.20	59.50	**87.30**
7	47.67	54.28	73.09	78.72	84.89	49.81	69.71	**96.74**	74.47
8	96.71	96.69	98.26	98.09	98.44	91.50	96.36	97.46	**98.56**
9	51.65	55.79	66.12	64.74	**87.93**	51.64	68.08	78.34	70.34
10	59.96	21.69	28.74	30.17	34.77	**74.59**	45.41	26.99	30.29
11	**93.21**	72.26	75.46	76.10	85.63	92.26	81.40	67.04	76.95
OA	86.59	87.35	92.61	91.45	91.79	86.29	89.08	92.22	**93.37**
	0.36	2.05	1.21	0.83	1.26	1.78	0.39	1.08	**1.02**
AA	76.14	71.46	80.32	80.52	81.51	78.03	78.52	81.07	**81.76**
	0.85	3.77	1.82	0.81	1.57	1.29	0.85	1.32	**1.71**
κ	82.43	83.42	90.19	88.67	89.06	82.04	85.59	90.58	**91.19**
	0.43	2.61	1.58	1.07	1.06	1.34	0.50	1.16	**1.32**

TABLE VIII

OA, AA, KAPPA COEFFICIENTS, AND CLASS-SPECIFIC ACCURACY(%) WITH DIFFERENT METHODS ON THE HOUSTON 2013 DATASET

Class No.	SVM [40]	CDCNN [41]	SSRN [42]	DBDA [43]	TBCNN [17]	EndNet [23]	MML [24]	HRWN [26]	MSNetSC
1	97.46	89.17	98.03	98.97	95.12	96.41	94.49	93.96	**99.69**
2	97.85	98.23	98.84	99.29	91.75	96.89	92.78	94.91	**99.88**
3	99.37	96.65	**100**	**100**	95.68	83.93	94.90	94.80	**100**
4	98.75	98.24	99.84	99.36	94.50	96.07	93.20	94.91	**99.97**
5	97.64	99.42	99.06	99.47	93.68	97.89	94.80	94.75	**100**
6	99.44	96.41	99.24	99.53	95.80	53.05	86.08	94.91	**100**
7	95.73	96.61	98.60	99.47	94.14	87.51	89.42	94.81	**99.93**
8	96.27	95.82	97.57	99.87	95.40	93.47	85.80	94.62	**99.97**
9	88.78	95.21	98.33	97.60	85.20	78.91	86.34	91.98	**99.58**
10	92.40	91.78	98.71	97.07	81.37	88.23	86.94	94.03	**99.91**
11	91.89	91.21	98.68	99.53	92.64	80.71	90.60	93.90	**99.98**
12	91.66	94.27	98.55	99.15	80.21	75.07	85.31	91.45	**99.71**
13	80.83	95.99	**99.62**	97.87	94.90	62.43	87.40	94.86	99.59
14	96.78	98.68	99.46	**100**	95.19	98.21	84.49	94.80	**100**
15	99.41	99.94	98.65	**100**	95.56	99.56	94.72	94.89	99.76
OA	94.98	95.05	98.57	99.05	90.81	87.06	90.14	94.19	**99.86**
	0.39	2.09	0.25	0.48	0.12	0.23	0.42	0.65	**0.08**
AA	94.95	95.84	98.88	99.15	92.08	85.89	89.82	94.24	**99.86**
	0.54	1.81	0.18	0.48	0.10	0.19	0.38	0.52	**0.09**
κ	94.56	94.63	98.54	98.97	90.35	85.24	89.73	94.12	**99.85**
	0.42	2.26	0.27	0.52	0.15	0.27	00.47	0.73	**0.09**

Furthermore, unlike other multimodality classification methods that roughly concatenate on the features from different modalities, the proposed MSNetSC model introduces the AFF module to dynamically calibrate the features from different modalities, thereby effectively fusing features for better multimodality data classification. With the same number of training

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TABLE IX
OA, AA, KAPPA COEFFICIENTS, AND CLASS-SPECIFIC ACCURACY(%) WITH DIFFERENT METHODS ON THE HOUSTON 2018 DATASET

Class No.	SVM [40]	CDCNN [41]	SSRN [42]	DBDA [43]	TBCNN [17]	EndNet [23]	MML [24]	HRWN [26]	MSNetSC
1	87.45	88.51	95.14	89.44	90.51	95.28	**95.99**	80.79	90.22
2	91.47	93.52	94.78	95.89	95.53	93.89	92.11	**98.24**	96.93
3	97.46	89.74	99.92	99.61	99.02	99.68	**100**	**100**	**100**
4	87.03	91.57	94.10	92.97	98.96	91.69	92.21	**99.80**	95.89
5	44.64	63.90	78.67	90.42	96.51	85.13	89.09	**98.49**	93.89
6	83.59	78.41	99.53	99.24	99.12	95.16	95.86	**99.81**	99.18
7	91.07	79.69	98.67	99.33	98.83	91.47	95.96	**100**	**100**
8	64.22	60.87	89.74	81.64	98.86	77.63	75.66	**99.85**	96.45
9	99.07	99.60	**99.78**	99.71	92.49	99.62	99.72	91.62	**99.78**
10	65.03	82.40	93.14	88.89	92.05	85.37	82.08	89.55	**96.23**
11	61.78	68.61	86.40	77.15	78.21	75.26	73.84	83.30	**89.77**
12	16.59	17.55	60.88	45.62	52.45	42.09	57.36	47.41	**62.83**
13	69.28	66.05	92.27	77.98	97.64	81.55	77.72	**98.95**	95.58
14	59.09	88.35	96.62	95.68	96.79	94.56	96.60	94.08	**99.44**
15	87.85	94.70	99.22	98.87	99.06	98.29	97.68	99.73	**99.98**
16	79.72	83.10	97.68	96.00	98.01	95.28	95.85	**99.84**	99.18
17	11.21	14.47	83.92	70.62	98.84	47.82	75.86	**100**	**100**
18	66.28	84.97	96.47	92.82	98.01	93.14	93.02	**99.67**	98.82
19	50.62	83.93	98.74	92.46	98.02	88.21	91.92	**99.82**	99.42
20	80.88	81.06	99.59	98.32	99.12	97.05	98.42	**100**	99.78
OA	81.00	84.61	95.55	91.93	93.21	91.46	90.62	93.17	**97.45**
	0.10	3.51	0.74	1.38	1.56	0.78	0.52	0.25	**0.16**
AA	69.72	75.55	92.76	89.13	93.90	86.41	88.85	94.05	**95.67**
	0.30	4.98	1.51	1.38	1.38	0.64	0.49	0.32	**0.22**
κ	75.95	80.09	94.10	89.35	91.37	88.75	87.66	91.04	**96.61**
	0.11	4.45	0.96	1.80	1.65	0.62	0.44	0.29	**0.28**

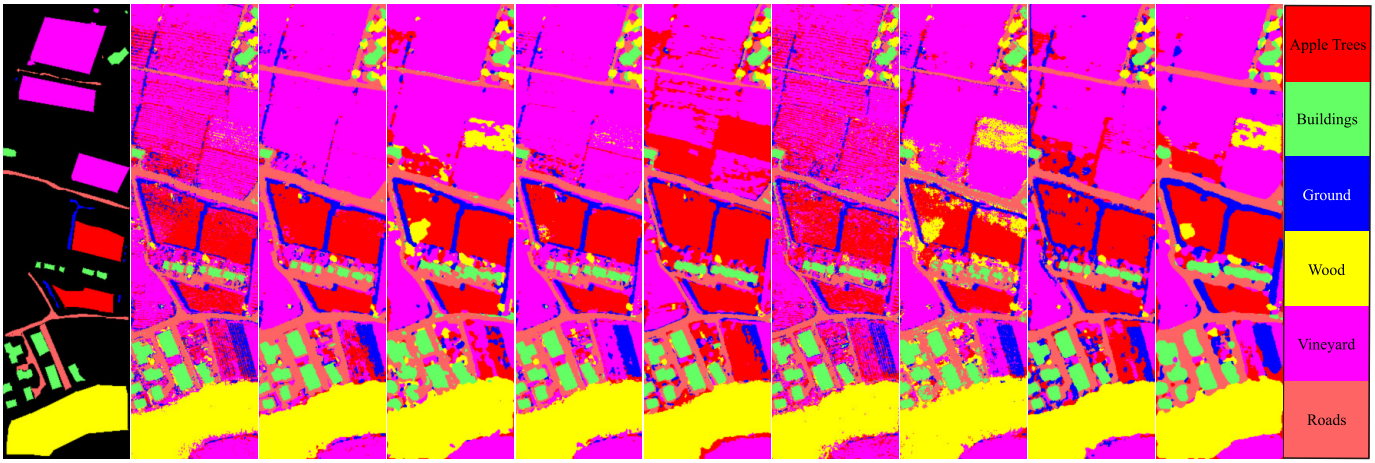

Fig. 9. Classification maps of different methods on the Trento dataset. (a) Ground-truth map. (b) SVM. (c) CDCNN. (d) SSRN. (e) DBDA. (f) TBCNN. (g) EndNet. (h) MML. (i) HRWN. (j) MSNetSC.

samples, the proposed method achieves the best classification accuracy in terms of OA, AA, kappa coefficient (κ), and individual class accuracy.

Last but not least, by comparing the accuracies of four different multimodality datasets, it can be found that the accuracies of the Houston 2018 dataset obtained by different methods are the lowest. The reason behind this phenomenon is that the structure of this dataset is more complex. However, by extracting the multiscale features from hyperspectral and LiDAR data, the classification accuracy of our proposed MSNetSC model can be significantly improved. The average OA of the MSNetSC reaches 97.45%, which is 1.90% higher than the SSRN model (i.e., 95.55%). Therefore, the multiscale features are of great importance for multimodality data classification, especially for datasets with complex structures.

2) Visual Comparisons: In addition to quantitatively evaluate the classification results using main evaluation coefficients, we also employ the classification maps for qualitative assessment. Figs. 9–12 show the classification maps with different classification models on four multimodality datasets. In these figures, different colors represent different land-cover classes, and ground-truth maps are also displayed to better observe classification results. When comparing the classification maps in (b)–(j) with ground-truth map in (a), we can find that the classification maps achieved by multimodality data classification models with deeper network structures have less noisy

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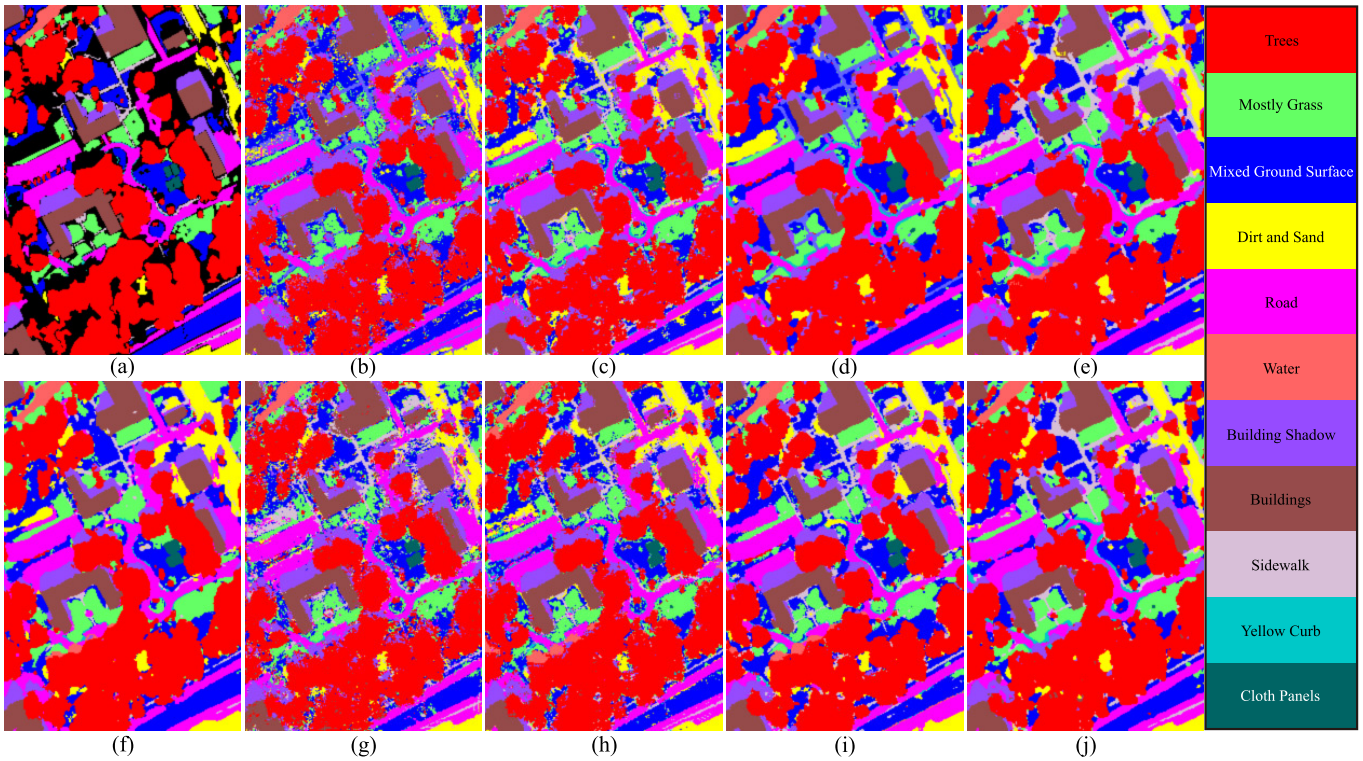

Fig. 10. Classification maps of different methods on the MUUFL Gulfport dataset. (a) Ground-truth map. (b) SVM. (c) CDCNN. (d) SSRN. (e) DBDA. (f) TBCNN. (g) EndNet. (h) MML. (i) HRWN. (j) MSNetSC.

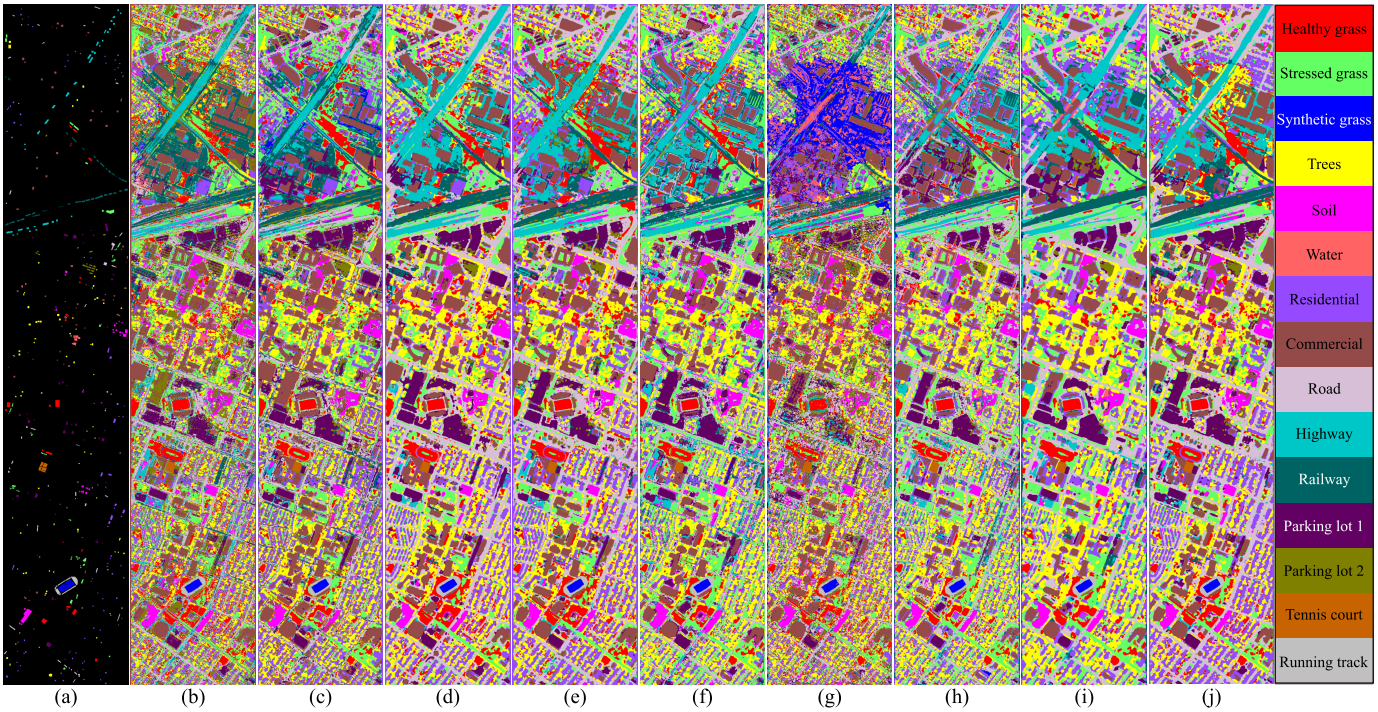

Fig. 11. Classification maps of different methods on the Houston 2013 dataset. (a) Ground-truth map. (b) SVM. (c) CDCNN. (d) SSRN. (e) DBDA. (f) TBCNN. (g) EndNet. (h) MML. (i) HRWN. (j) MSNetSC.

pixels. Also, the classification maps obtained by our MSNetSC method are more homogeneous, which have the fewest noisy pixels. More specifically, take the Houston 2018 dataset for example, and there are many noisy pixels in the Nonresidential buildings class obtained by other comparative methods, but our method can obtain the most homogeneous classification map.

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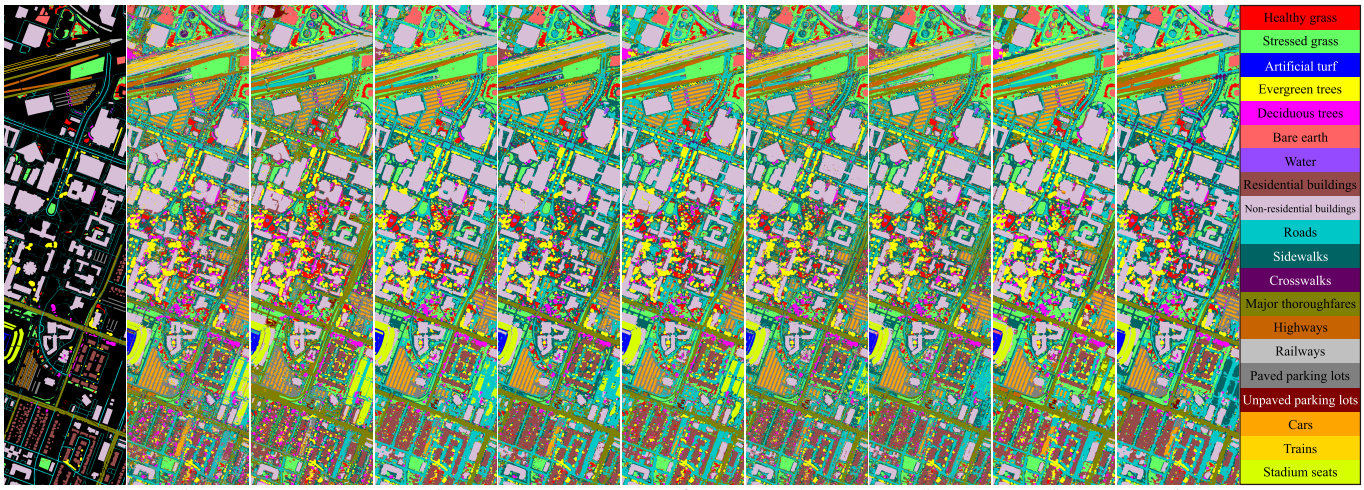

Fig. 12. Classification maps of different methods on the Houston 2018 dataset. (a) Ground-truth map. (b) SVM. (c) CDCNN. (d) SSRN. (e) DBDA. (f) TBCNN. (g) EndNet. (h) MML. (i) HRWN. (j) MSNetSC.

TABLE X
AVERAGE OA AND VARIANCE (%) OF THE MSNETSC MODEL WITH DIFFERENT DOWN-SAMPLING RATES ON FOUR MULTIMODALITY DATASETS

Rate	Trento	MUUFL Gulfport	Houston 2013	Houston 2018
1	99.57 ±0.18	92.70 ±1.76	99.78 ±0.20	97.29 ±0.31
2	**99.66 ±0.08**	93.30 ±0.61	99.70 ±0.24	**97.45 ±0.16**
3	99.60 ±0.14	93.16 ±0.75	**99.86 ±0.08**	97.34 ±0.46
4	99.65 ±0.09	**93.37 ±1.02**	99.80 ±0.13	97.17 ±0.22
5	99.57 ±0.14	92.99 ±0.49	99.79 ±0.11	97.31 ±0.27

This is because our method can extract more discriminative features from multimodality data and adaptively calibrate these features to obtain better classification results. Also, for the Stadium seats in the Houston 2018 dataset, they distribute at different elevations, but our method can effectively fuse LiDAR-derived DSM and get the most reasonable classification results.

D. Parameter Sensitivity Analysis

1) Down-Sampling Rate: In the self-calibrated convolutional layer, the field-of-view is explicitly expanded through internal communications within the convolution operation. After the down-sampling operation, the embeddings in the latent space are employed as references to guide the feature transformation process in the original feature space because of their large field-of-view. Thus, the down-sampling rate determines the size of internal field-of-view and then influences the discriminative capability of extracted features to a certain degree. We set the down-sampling rate at $\{1, 2, 3, 4, 5\}$, and Table X shows the average OA with variance of the proposed method on four multimodality datasets. From the results, we can see that the classification accuracy generally increases and then decreases with the increase of the down-sampling rate. Also, the classification accuracies of four datasets are affected by this parameter to varying degrees. According to the experimental results, the optimal down-sampling rates of four multimodality datasets are set as 2, 4, 3, and 2.

2) Spatial Size: Since the inputs of MSNetSC method are image blocks from hyperspectral and LiDAR data, the neighborhood size will affect the classification performance to a certain degree. Also, we set the neighborhood size from a candidate set $\{3, 5, 7, 9, 11, 13, 15\}$ to evaluate the effect of this parameter. Fig. 13 reports the overall accuracies on four multimodality datasets with different spatial sizes. In this figure, two independent horizontal lines represent the overall range of the classification results, and the two edges of the box denote the upper quartile and lower quartile, respectively. The horizontal line in box refers to the median value, and the ♦ denotes abnormal outliers. It can be found that the spatial size has different influences on four groups of datasets in terms of classification accuracy. The overall accuracies of MUUFL Gulfport and Houston 2018 datasets are greatly affected by this parameter, and the variations of average OA are more than 3% and 2%, respectively. For the Trento and Houston 2013 datasets, the classification accuracies are relatively stable when the spatial size changes. Based on experimental results with different neighborhood sizes, the reasonable spatial sizes for four datasets are 9, 5, 7, and 11.

3) Learning Rate: The learning rate controls the step of gradient descent in the MSNetSC model, which can also affect the classification performance. The candidate set of learning rate is $\{0.0001, 0.0002, 0.0003, 0.0005, 0.0008, 0.001, 0.005, 0.01\}$, and we test the effects of this parameter on the overall accuracies. Fig. 14 reports overall accuracies with different learning rates on four multimodality datasets. From this figure, we can see that a smaller learning rate has a relatively stable classification accuracy and a bigger learning rate will result in a larger variance in the classification accuracy. Also, this parameter has an obvious impact on the MUUFL Gulfport and Houston 2013 datasets but less on the

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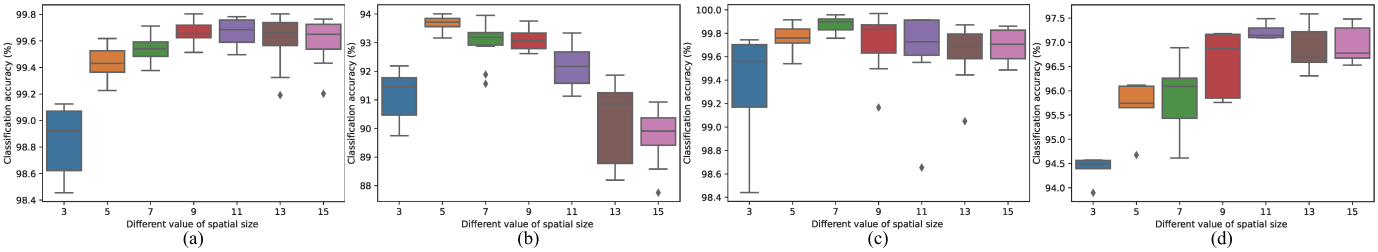

Fig. 13. Box plot of overall accuracies with different spatial sizes on four multimodality datasets. (a) Trento dataset. (b) MUUFL Gulfport dataset. (c) Houston 2013 dataset. (d) Houston 2018 dataset.

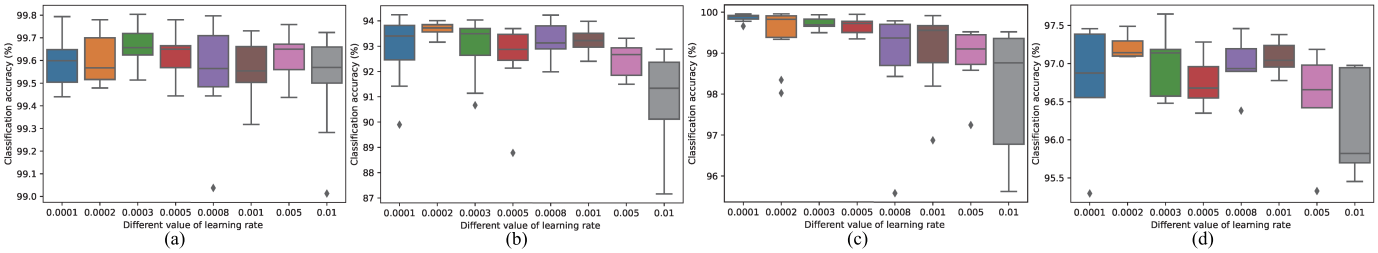

Fig. 14. Box plot of overall accuracies with different learning rates on four multimodality datasets. (a) Trento dataset. (b) MUUFL Gulfport dataset. (c) Houston 2013 dataset. (d) Houston 2018 dataset.

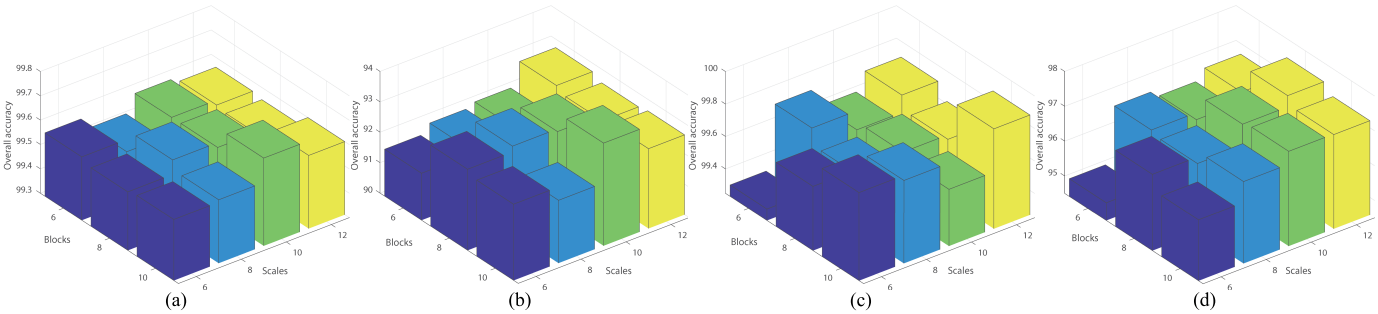

Fig. 15. Bar chart of average overall accuracies with different network structures on four multimodality datasets. (a) Trento dataset. (b) MUUFL Gulfport dataset. (c) Houston 2013 dataset. (d) Houston 2018 dataset.

other two datasets. According to the experimental results on four groups of datasets, the reasonable learning rates are set as 0.0003, 0.0002, 0.0001, and 0.0002.

4) Feature Extraction Structure: In the MSNetSC model, different feature extraction structures can extract features of different scales and layers, which affects the representation capability of extracted multiscale features. The scale of the hierarchical residual block and the number of feature maps in each subgroup determine the structure of the feature extraction network, and we denote these two hyperparameters as scale and block. In order to evaluate the classification accuracy of MSNetSC model with different feature extraction network structures, we evaluate the influence of these two hyperparameters using a cross-validation strategy. The candidate sets of these two hyperparameters are $\{6, 8, 10, 12\}$ and $\{6, 8, 10\}$. The average overall accuracies using different feature structures on four datasets are shown in Fig. 15. Based on the experimental results, we can learn that the classification accuracy is generally lower when the numbers of scales and kernels are lower, especially for the Houston 2018 dataset. Since the scale parameter directly determines the scales of extracted features, higher scales can effectively improve the classification accuracy, especially for more complex datasets. Also, the optimal parameter combinations of four multimodality datasets are selected as $\{10, 10\}$, $\{10, 10\}$, $\{12, 10\}$, and $\{12, 8\}$.

5) Analysis on Computation Cost: To quantitatively analyze the computation cost of different methods, we report their computation time on four multimodality datasets in Table XI. The training time is mainly affected by the number of training samples and the network parameters, and the number of training samples is the same in each group of experiments, so the computation cost can be effectively measured by training time. Since the validation set influences iterations in different models, we employ the average training time in an epoch for a fair comparison. From this table, we can observe that deeper classification models on more complex datasets need more training time. Also, our training procedure of MSNetSC takes more time, which is because of the introduction of attention modules to our method. Nevertheless, the increase in time is acceptable, and the improvement of classification accuracy in our proposed method is obvious.

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TABLE XI
COMPUTATION TIME WITH DIFFERENT METHODS ON FOUR MULTIMODALITY DATASETS AND TRAIN TIME RECORDS THE TIME OF EACH EPOCH

Data Set	Time(s)	CDCNN	SSRN	DBDA	TBCNN	EndNet	MML	HRWN	MSNetSC
Trento	Train	1.02	1.18	0.65	1.43	1.37	1.26	1.46	2.16
	Test	5.44	7.83	4.76	13.24	10.35	9.36	14.95	16.85
MUUFL Gulfport	Train	3.40	4.02	2.26	4.36	3.84	4.21	5.02	6.95
	Test	10.62	14.55	12.43	21.42	18.90	25.04	34.01	39.82
Houston 2013	Train	2.32	3.98	3.45	4.53	2.56	4.14	5.78	8.66
	Test	8.97	12.58	11.35	12.24	10.82	14.08	16.86	18.10
Houston 2018	Train	12.35	20.16	24.89	23.64	16.07	20.03	21.62	22.18
	Test	32.97	76.61	56.02	62.74	49.38	58.58	65.34	77.98

TABLE XII
ABLATION STUDY ON FOUR MULTIMODALITY DATASETS

	Trento			MUUFL Gulfport			Houston 2013			Houston 2018		
	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ
MSNetSC	99.66	99.14	99.55	93.37	81.76	91.19	99.86	99.86	99.85	97.45	95.67	96.61
	0.08	0.28	0.10	1.02	1.71	1.32	0.08	0.09	0.09	0.16	0.22	0.28
MSNet	99.65	99.09	99.53	92.91	81.07	90.60	99.69	99.71	99.66	97.26	94.75	96.36
	0.14	0.36	0.19	1.02	1.47	1.33	0.23	0.29	0.25	0.18	0.30	0.23
MSNetSC (Without Self-attention)	99.61	99.19	99.47	92.90	81.34	90.60	99.75	99.79	99.73	97.22	94.96	96.30
	0.13	0.28	0.18	1.44	1.73	1.86	0.17	0.13	0.19	0.25	0.75	0.19
MSNetSC (Concatenation)	99.66	99.02	99.55	92.70	80.85	90.33	99.72	99.74	99.70	96.76	95.07	95.69
	0.09	0.38	0.13	1.59	1.14	1.76	0.19	0.16	0.20	0.36	0.66	0.48
MSNetSC (Spectral+Spatial)	98.98	97.73	98.64	92.10	80.55	89.55	99.67	99.68	99.64	96.14	94.12	94.88
	0.16	0.54	0.21	1.80	1.81	1.23	0.25	0.22	0.27	0.62	0.54	0.81

E. Ablation Study

We further carried out different ablation studies to highlight an individual aspect of our model. Because the proposed MSNetSC model benefits from a combination of several aspects, we evaluate our model's performance by iteratively removing each of these aspects, and the ablation results are reported in Table XII. In this table, the average overall accuracies with the corresponding variances are employed to evaluate the classification performance.

In order to test the contribution of self-calibrated convolution to the MSNetSC, we replace the self-calibrated convolution with vanilla convolution in our model, and we name this new model as MSNet. It can be seen from Table XI that the classification accuracies using MSNet on four datasets are generally lower than those obtained by MSNetSC, which proves the effectiveness of the self-calibrated convolution in MSNetSC.

The self-attention module in MSNetSC can calibrate the weights of extracted multiscale features, which can further enhance the discriminative capability of the features. With the purpose of verifying the effectiveness of self-attention module, we remove the self-attention modules from our method, and this new model is denoted as MSNetSC (without self-attention). The classification accuracies of this new model are also listed in Table XI. Through the comparative experiments, we can observe that the self-attention module can improve the classification accuracy to a certain extent.

To validate the effectiveness of the AFF module in our proposed method, we concatenate three types of multiscale features for collaborative classification, and we name this new model as MSNetSC (Concatenation). By comparing the experimental results in this table, we can come to the conclusion that using the nonlinear AFF module can effectively fuse multiple features for better classification results.

In addition, the importance of elevation information of LiDAR-derived rasterized DSM has also been proved since the hyperspectral and LiDAR data-based classification model gets better accuracy than the hyperspectral data-based model, denoted as MSNetSC (Spectral+Spatial), on all four multimodality datasets. Through the above ablation study, we can see that in the absence of even anyone of these aspects, the model tends to underperform, and thus, our proposed MSNetSC model benefits from a combination of several aspects.

F. Discussion

When using multimodality data for land-cover classification, traditional deep learning models have difficulties in extracting multiscale features and fusing multimodality information, which will affect the classification performance to a certain extent. To address these problems, we propose the MSNetSC, and this model can extract multiscale features from multimodality data (i.e., hyperspectral and LiDAR data) and effectively fuse these multimodality features for collaborative classification. In four groups of multimodality data classification experiments, the MSNetSC method achieves better classification results compared with other state-of-the-art deep learning models. The main reasons can be summarized as the following two aspects.

First, the multiscale feature extraction ability is vital important for multimodality data collaborative classification. The proposed model simultaneously utilized a hierarchical residual

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network and self-calibrated convolution to extract features with different receptive fields, so as to learn the multiscale information. Also, the extracted features with different scales make contributions to multimodality data classification, and especially for more complex images, such as the Houston 2018 dataset, the classification accuracy will improve significantly. The comparative experiments and ablation study confirm the importance of multiscale features for multimodality data joint classification.

Second, the attention-based modules can calibrate the weights of features for better classification performance. In our proposed method, the self-attention modules adaptively set different weights for features with different scales during the training process, which can improve useful information and suppress unfavorable features. Also, the AFF module assigns reasonable weights for features from different modalities to effectively utilize multimodality information for land-cover classification. Also, the importance of these two types of attention-based modules has been verified by comparative experiments and ablation study.

IV. Conclusion

In this study, we propose a novel MSNetSC for multimodality data (i.e., hyperspectral and LiDAR data) collaborative classification, which has three advantages. The first one is that the proposed network simultaneously utilizes hierarchical residual structure and self-calibrated convolution to extract discriminative features of different scales, thereby learning multiscale information for classification. The second one is that the self-attention modules are employed to adaptively calibrate the weights of extracted multiscale features, and the third one is the AFF module, which presents a nonlinear approach for feature fusion in a context-awareness way. Therefore, the novel MSNetSC model can extract more complete and discriminative features and effectively fuse multimodality information for collaborative classification. The residual learning structure and BN can ensure training efficiency during the training process. The performance of MSNetSC has been verified on four benchmark multimodality datasets compared with other state-of-the-art methods, and the comparative results have confirmed the superiority of the proposed method.

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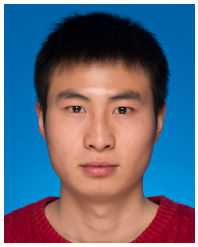

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